

Mohit Kumar Wankhade

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Summary

AI/ML and Software Engineer focused on building production-ready GenAI applications, RAG systems, backend APIs, and machine learning solutions. Experienced with FastAPI, LLM APIs, LangChain, vector search, NLP, deep learning, databases, Dockerized deployments, monitoring, and cloud-based application delivery.

Technical Skills

Programming: Python, C++, JavaScript, SQL

Generative AI: LLMs, RAG, Prompt Engineering, Embeddings, Vector Search, LangChain, Sentence Transformers

Retrieval: FAISS, BM25, Hybrid Search, Cross-Encoder Reranking, Source Attribution, Context Building

Backend/SWE: FastAPI, REST APIs, JWT Authentication, API Design, OOP, Debugging, Logging, Testing

Frontend: React, Vite, Tailwind CSS, Axios, React Router, HTML, CSS

Databases & Cache: MongoDB, MongoDB Atlas, Redis, MySQL

ML/DL: PyTorch, TensorFlow, Scikit-learn, NLP, Computer Vision, Transformers, Multimodal Learning, Classification

DevOps/Cloud: Docker, Docker Compose, Nginx, Prometheus, Grafana, AWS, Oracle Cloud, Git, GitHub

Experience

Tata Consultancy Services (TCS)

May 2025 – Present

Software Engineer – GenAI Applications

Pune, Maharashtra

- Worked on GenAI chatbot workflows involving LLM APIs, semantic retrieval, REST APIs, and backend services for contextual query handling.
- Designed FastAPI-based API modules with request validation, exception handling, structured logging, and reusable service-layer logic.
- Integrated MongoDB-based conversation memory to support context-aware interactions, persistent chat history, and multi-turn conversations.
- Performed monitoring, API testing, debugging, incident analysis, and performance validation to improve application stability and response reliability.

Indian Institute of Technology (IIT) Patna

Jan 2024 – May 2024

NLP Research Intern under Dr. Sriparna Saha

Patna, Bihar

- Developed a multimodal financial complaint identification framework using text, image, sentiment, emotion, severity, and aspect-level signals.
- Worked on financial social media data covering 6,439 annotated samples, 5 financial domains, and 10 aspect categories.
- Built transformer and deep learning pipelines using RoBERTa, ResNet-152, BiGRU, self-attention, and BART, achieving up to 93% complaint classification accuracy.

Projects

MW GenAI Chatbot – Production RAG Assistant

Live Demo | [GitHub](#)

FastAPI, React, LangChain, FAISS, BM25, MongoDB Atlas, Redis, Docker, Nginx, Oracle Cloud

- Built and deployed a full-stack RAG chatbot supporting PDF upload, GitHub repository indexing, streaming responses, and source-grounded answers.
- Implemented hybrid retrieval using FAISS dense search, BM25 sparse search, result fusion, and Cross-Encoder reranking to improve context relevance.
- Added JWT authentication, protected routes, MongoDB conversation memory, Redis response caching, and source attribution for PDF pages and GitHub files.
- Containerized and deployed the complete stack using Docker Compose, Nginx reverse proxy, Prometheus metrics, Grafana dashboards, health checks, and Oracle Cloud VM.

Multimodal Financial Complaint Identification Framework

[GitHub](#)

Python, PyTorch, RoBERTa, ResNet-152, BiGRU, BART, NLP, Computer Vision

- Designed an aspect-based multimodal complaint identification framework using textual, visual, sentiment, emotion, and severity features.
- Processed 6,439 financial social media samples across 5 financial domains and 10 aspect categories with complaint/non-complaint annotations.
- Combined RoBERTa text embeddings with ResNet-152 image features and self-attention layers, achieving 93% accuracy/micro-F1 and 91% macro-F1.

Education

Madhav Institute of Technology and Science, Gwalior

2020 – 2024

Bachelor of Technology in Artificial Intelligence and Robotics

CGPA: 8.07/10

Achievements

- GATE Qualified in Data Science & AI and Computer Science
- Solved 523+ problems on LeetCode with a 430-day coding streak, 365+ active days, and multiple coding badges
- Certifications: Supervised Machine Learning and Advanced Learning Algorithms – Coursera, Stanford University